

Module-IV
Data Collection & Interpretation and Report Writing

The **primary data** are those which are collected afresh and for the first time, and thus happen to be original in character.

The **secondary data**, on the other hand, are those which have already been collected by someone else and which have already been passed through the statistical process.

COLLECTION OF PRIMARY DATA:

There are several methods of collecting primary data, particularly in surveys and descriptive research. Important ones are:

- i. observation method,
- ii. interview method,
- iii. through questionnaires

Observation Method:

The observation method is used in studies relating to behavioral sciences.

Observation becomes a scientific tool and the method of data collection for the researcher, when it serves a formulated research purpose, is systematically planned and recorded and is subjected to checks and controls on validity and reliability.

Advantages

- The information obtained under this method relates to what is currently happening
- This method is independent of respondents' willingness to respond

Limitations:

- it is an expensive method.
- the information provided by this method is very limited.
- sometimes unforeseen factors may interfere with the observational task.

Merits of the participant type of observation:

- The researcher is enabled to record the natural behavior of the group.
- The researcher can even gather information which could not easily be obtained

Controlled and Uncontrolled observation:

- If the observation takes place in the natural setting, it may be termed as uncontrolled observation
- when observation takes place according to definite pre-arranged plans, involving experimental procedure, the same is then termed controlled observation.

Interview Method:

The interview method of collecting data involves presentation of oral-verbal stimuli and reply in terms of oral-verbal responses.

This method can be used through personal interviews and, if possible, through telephone interviews.

(a) Personal interviews:

Personal interview method requires a person known as the interviewer asking questions generally in a face-to-face contact to the other person or persons.

This sort of interview may be in the form of direct personal investigation, or it may be indirect oral investigation.

In the case of direct personal investigation, the interviewer has to collect the information personally from the sources concerned. He has to be on the spot and has to meet people from whom data have to be collected.

But in certain cases it may not be possible or worthwhile to contact directly the persons concerned or on account of the extensive scope of enquiry, an indirect oral examination can be conducted under which the interviewer has to cross-examine other persons who are supposed to have knowledge about the problem under investigation and the information, obtained is recorded.

Merits of the interview method:

- (i) More information and that too in greater depth can be obtained.
- (ii) There is greater flexibility, as the opportunity to restructure questions is always there, specially in case of unstructured interviews.
- (iii) Personal information can also be obtained easily under this method.
- (iv) Non-response generally remains very low.
- (v) The interviewer can usually control which person will answer the questions.
- (vi) The language of the interview can be adopted to the ability or educational level of the person interviewed and as such misinterpretations concerning questions can be avoided.

Weaknesses of the interview method:

- It is a very expensive method
- people in high income groups may not be easily approachable
- This method is relatively more-time-consuming
- There remains the possibility of the bias of interviewer as well as that of the respondent
- It is required for selecting, training and supervising the field-staff is more complex

For successful implementation of the interview method:

- interviewers should be carefully selected, trained and briefed.
- They should be honest, sincere, hardworking, impartial and must possess the technical competence and necessary practical experience.
- Occasional field checks should be made to ensure that interviewers are neither cheating, nor deviating from instructions given to them for performing their job efficiently

(b) Telephone interviews:

This method of collecting information consists in contacting respondents on telephone itself. It is not a very widely used method, but plays important part in industrial surveys, particularly in developed regions.

The chief merits of such a system are:

- Flexible
- Faster
- Cheaper than personal interview methods
- Response can be recorded
- No field staff is required
- More samples can be obtained

Demerits of telephonic interviews:

- Little time is given to respondents
- Surveys are restricted to respondents who have telephone facilities.
- Possibility of the bias of the interviewer is relatively more.
- Questions have to be short and to the point

COLLECTION OF DATA THROUGH QUESTIONNAIRES:

This method of data collection is quite popular, particularly in case of big enquiries. It is being adopted by private individuals, research workers, private and public organizations and even by governments.

Merits:

- Low cost
- Respondents will get adequate time
- Respondents, who are not easily approachable, can also be reached conveniently.
- Large samples can be made use of and thus the results can be made more dependable and reliable.

Demerits:

- Low rate of return
- It can be used only when respondents are educated
- The control over questionnaire may be lost once it is sent
- It is difficult to know whether willing respondents are truly representative.
- This method is likely to be the slowest of all.

MEANING OF INTERPRETATION:

Interpretation refers to the task of drawing inferences from the collected facts after an analytical and/or experimental study.

In fact, it is a search for broader meaning of research findings.

The task of interpretation has two major aspects viz.,

(i) the effort to establish continuity in research through linking the results of a given study with those of another,

(ii) the establishment of some explanatory concepts.

Why INTERPRETATION?

- Continuity in research can be maintained.
- to understand the real significance of his research findings

TECHNIQUES OF INTERPRETATION:

- Extraneous information, must be considered while interpreting the final results of research study,
- It is advisable to consult someone having insight into the study
- Researcher must consider all relevant factors affecting the problem.

DIFFERENT STEPS IN WRITING REPORT:

- logical analysis of the subject-matter;
- preparation of the final outline;
- preparation of the rough draft;
- rewriting and polishing;
- preparation of the final bibliography; and
- writing the final draft.

LAYOUT OF THE RESEARCH REPORT:

The layout of the report means as to what the research report should contain.

A comprehensive layout of the research report should comprise of:

- preliminary pages.
- the main text.
- the end matter.

A. Preliminary pages:

- In its preliminary pages the report should carry a title and date, followed by acknowledgements in the form of 'Preface' or 'Foreword'.
- Then there should be a table of contents followed by list of tables and illustrations so that the decision-maker or anybody interested in reading the report can easily locate the required information in the report.

B. Main Text:

The main text of the report should have the following sections:

Introduction

Review of Literature

Methodology

Results and discussion

Conclusions

Scope for future work

C. End Matter:

- At the end of the report, appendices should be enlisted in respect of all technical data such as questionnaires, sample information, mathematical derivations and the like ones.
- Bibliography of sources consulted should also be given.

MECHANICS OF WRITING A RESEARCH REPORT:

Size and physical design:

The manuscript should be written on unruled paper 8 ½ " × 11" in size.

A margin of at least one and one-half inches should be allowed at the left hand and of at least half an inch at the right hand of the paper.

There should also be one-inch margins, top and bottom. All typing should be double-spaced on one side of the page only.

Procedure:

Various steps in writing the report should be strictly adhered.

Layout:

Keeping in view the objective and nature of the problem, the layout of the report should be thought of and decided and accordingly adopted.

Treatment of quotations:

Quotations should be placed in quotation marks and double spaced

The footnotes:

Footnotes are placed at the bottom of the page on which the reference or quotation which they identify or supplement ends.

The final draft:

Revising and rewriting the rough draft of the report should be done with great care before writing the final draft. For the purpose, the researcher should put to himself questions like: Are the sentences written in the report clear? Are they grammatically correct? Do they say what is meant? Do the various points incorporated in the report fit together logically?

Bibliography and References:

Bibliography should be prepared